

PROFESSIONAL SUMMARY

Ph.D. candidate in Data Science specializing in clinical informatics and AI-driven healthcare applications. Expertise in medical knowledge representation, ontology engineering, Large Language Models (LLMs), and clinical terminology standards including, SNOMED CT, ICD-10, and UMLS. Passionate about developing trustworthy AI systems that improve healthcare interoperability, clinical decision support, and biomedical knowledge management

EDUCATION

|                                                                                                |                           |
|------------------------------------------------------------------------------------------------|---------------------------|
| <b>Ph.D.</b> , New Jersey Institute of Technology                                              | September 2023 – Present  |
| Major: Data Science                                                                            | GPA: 3.7/4.00             |
| Coursework: Applied Statistics, Introduction to Big Data, Machine Learning.                    |                           |
| <b>Master of Science</b> , New Jersey Institute of Technology                                  | September 2021 – May 2023 |
| Major: Computer Science                                                                        | GPA: 3.8/4.00             |
| Coursework: Java Programming, Database Management System Design, Operating Systems.            |                           |
| <b>Bachelor of Engineering</b> , University of Mumbai                                          | May 2017 – May 2021       |
| Major: Computer Engineering                                                                    | GPA: 8.25/10.00           |
| Coursework: Data Structures, Analysis of Algorithms, Computer Networks, Compiler Construction. |                           |

SKILLS

|                            |                                                                                 |
|----------------------------|---------------------------------------------------------------------------------|
| Programming                | Python, Java, Kotlin, JavaScript, TypeScript, Chapel, C/C++                     |
| Web Technologies           | HTML, CSS, Angular                                                              |
| Data Science               | pandas, NumPy, scikit-learn, matplotlib, Jupyter, Streamlit                     |
| Medical Informatics        | UMLS, SNOMED CT, ICD-10, Medical Ontologies, Incremental Ontology Expansion     |
| AI and Machine Learning    | LLMs, Retrieval-Augmented Generation, Machine Learning, Agentic Workflows (n8n) |
| Databases                  | Database Design, PL/SQL, Data Modeling, Query Optimization                      |
| Development Tools          | Git, GitHub, Jenkins, CI/CD, Jira, Argo, LaTeX                                  |
| High-Performance Computing | Arkouda, Chapel, Parallel Computing, Distributed Data Analysis                  |

TEACHING EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>Adjunct Instructor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | September 2024 - Present |
| <b>New Jersey Institute of Technology</b> , New Jersey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |
| <b>Fall 2024 and Summer 2025</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |
| <ul style="list-style-type: none"><li>Teach CS 331: <b>Database System Design and Management</b>, covering key concepts such as database design, SQL programming, normalization, and transaction management, providing both theoretical and practical knowledge.</li><li>Develop comprehensive course materials and assessments, ensuring students gain a robust understanding of database systems and are well-prepared for real-world applications in the domain.</li></ul>                                                                                |                          |
| <b>Teaching Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | September 2023 - Present |
| <b>New Jersey Institute of Technology</b> , New Jersey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                          |
| <ul style="list-style-type: none"><li>Teaching Assistant for CS 632: <b>Advanced Database System Design</b>, CS 331: <b>Database System Design and Management</b>, DS 636: <b>Data Analytics with R programming</b>, DS 644: <b>Introduction to Big Data</b>, and CS 434: <b>Advanced Database Systems</b></li><li>Conduct weekly consultations during dedicated office hours to address graduate student questions and provide clarifications.</li><li>Assess and grade student assignments and homework, providing timely and accurate feedback.</li></ul> |                          |

WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Graduate Researcher</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | October 2021 - August 2023 |
| <b>New Jersey Institute of Technology</b> , New Jersey                                                                                                                                                                                                                                                                                                                                                                                                       |                            |
| <ul style="list-style-type: none"><li>Conducted research on <b>Arkouda</b> – a software package that allows a user to interactively issue <b>massive parallel computations</b> on distributed data using functions and syntax that mimic NumPy, the underlying computational library used in most Python data science workflows.</li><li>Evaluated user experience and performed performance benchmarking against existing data science workflows.</li></ul> |                            |
| <b>Technology Engineering Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | June 2022 - August 2022    |
| <b>Forbes Media LLC</b> , New Jersey                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |
| <ul style="list-style-type: none"><li>Improved the User Interface of the Content Management System Bertie by creating Angular components per the Product Design Team – <b>improved user interaction with the CMS by seventy percent</b>.</li></ul>                                                                                                                                                                                                           |                            |

- Contributed to the **pilot team responsible for the iOS and Android Development** of the Forbes Mobile Application.
- Developed the architecture for Multivendor Call to Action features to **increase the customer conversion rate by fifty percent**.

Network Operations Center Intern

GPX India Pvt. Ltd (now Equinix), Mumbai, India

December 2019 - December 2019

- Worked with the Network Operations Center (NOC) and improved the workflow of Building Management System (BMS) by visualizing the existing Chiller Plants’ data using Python.
- Improved data-handling efficiency with the use of Python scripts saving six to eight hours of manual scraping of data in CSV files generated by the client data of the Data Center.

PUBLICATIONS

Conference Presentations

1. **Naren Khatwani**, Navya Martin Kollapally, Lijing Wang, James Geller. [Evaluating RAG and Non-RAG Pipelines for Concept Discovery in Environmental Health Ontologies](#). In *AMIA 2026 Informatics Summit*, 2026.
2. **Naren Khatwani**, Lijing Wang, Shmuel T. Klein, James Geller. [Convergence to steady state in LLM-Generated Ontological Concepts](#) . In *MIE 2026 - Medical Informatics Europe*”, 2026.
3. **Naren Khatwani**, Lijing Wang, James Geller. [To what Degree can LLMs Support Medical Informatics Research? Examining the Interplay of Research Support LLMs with LLM Critics](#). In *AMIA 2025 Annual Symposium*, 2025.
4. **Naren Khatwani**, Lijing Wang, James Geller. [A Concept Utility Framework For Incremental Ontology Expansion](#). In *MCCSIS 2025 - 17th International Conference on e-Health*, 2025.
5. **Naren Khatwani**, James Geller. [Population Group 2.0: Bringing the UMLS Semantic Network up to speed](#). In *MIE 2025 - "Intelligent health systems – From technology to data and knowledge"*, 2025.
6. Richard Joseph, **Naren Khatwani**, Rahul Sohandani, Raghav Potdar, and Adithya Shrivastava. [Food Aayush: Identification of Food and Oils Quality](#). In *2022 Springer - Smart Innovation, Systems and Technologies*, 2022.
7. Anish Adnani, **Naren Khatwani**, Supriya Patil. [Attenuation of a Pandemic Disease Spread](#). In *2020 International Research Journal of Engineering and Technology (IRJET)*, 2020.

Poster Presentation

1. **Naren Khatwani**, Nghia Bui, James Geller, and Lijing Wang. [Human-AI Agreement in Concept Evaluation: A Benchmarking Framework to Support Incremental Ontology Expansion..](#) In *2026 AMIA Annual Symposium*, 2026 (Accepted).

Invited Talks and Workshops

1. **Naren Khatwani**, James Geller. [A Concept Utility Framework for Incremental Ontology Expansion](#). Invited Seminar, *Joint George Washington University (GWU) and Washington DC VA Informatics Seminar*, 2025.
2. Oliver Alvarado Rodriguez, **Naren Khatwani**, Zhihui Du, and David Bader. [Interactive Large-Scale Data and Graph Analytics](#). Workshop Presentation, *Principles and Practice of Parallel Programming (PPoPP)*, 2023.

PROJECTS

[Analysis of Physiological Responses of Game Players](#)

Independent Study

2024

- This project worked on players’ physiological responses during League of Legends matches, specifically focusing on how intense gaming moments, such as kills, influence players’ stress levels. By measuring Galvanic Skin Response (GSR) and heart rate variations, I found patterns that indicated heightened emotional and cognitive stress during gameplay.

[Global Food Prices Data Analysis Using Parallel Computing](#)

Graduate Research Project

2022

- This Data Analysis project aimed at the performance comparison between Arkouda (a software package that allows a user to interactively issue massive parallel computations on distributed data) and Pandas by analyzing the open-source Global Food Price - dataset with more than 700K rows.
- The work done was under the domain of **High-Performance Parallel Computing** which is estimated to improve the limit of existing Pandas library by **four hundred percent**.

[Food AAYUSH](#)

Final Year Undergraduate Engineering Project

2021

- The project aimed to build an application for the analysis of the quality of food and to detect repeated frying on cooking oils based on the visual properties of the food items and the pH value of the oils.
- The analysis of the nutritional value of food items was done using a python-based framework known as Streamlit and the oil quality was analyzed via a **portable pH sensor and ESP32 Microcontroller**.

- The outcome of the project **helped youngsters and inexperienced vegetable buyers** determine the food and oil quality.

Google Play Store Application Analysis

2019

Data Analytics Project

- Analyzed Google Play Store application data to identify patterns in downloads, usage, ratings, and device compatibility.
- Applied sentiment analysis to user reviews to examine their relationship with application popularity and download trends.

ACCOMPLISHMENTS

Github Universe’24 Travel Award, GitHub, San Francisco, USA

2024

- GitHub Universe is the global, annual event for the developer community to hear from industry leaders and observe interactive sessions on AI, security and developer experience.

Github Universe’23 Travel Award, GitHub, San Francisco, USA

2023

- Actively participated in beta testing GitHub Copilot, explored CodeQL for advanced code analysis, and attended specialized demonstrations focused on preventing secret sharing leaks.

Reviewer’s Choice Award, IEMAICLOUD, Springer, UK

2021

- Our project “*FoodAayush*” was honored with the Reviewer’s Choice Award at IEMAI Cloud. Our findings and contributions were subsequently published in a [Springer book](#).

Best Technical Paper, Institute of Engineers, India

2019

- Our team was awarded the *Best Technical Paper* at the *Institute of Engineers* for our innovative approach in predicting structural lifespan. By analyzing construction parameters, our system aids civil engineers in enhancing construction efficiency and sustainability.

ACADEMIC SERVICE

Reviewer

2026

- International Joint Conference on Artificial Intelligence (IJCAI 2026) – AI for Healthcare Track
- EFMI Special Topic Conference (STC) 2026 - AI in Clinical Decision Making and Education

VOLUNTEERING

GitHub Campus Expert

May 2020 - Present

GitHub

- GitHub Campus Experts are student leaders that strive to build diverse and inclusive communities to learn skills, collaborate on projects, and share their experiences.
- Founded the NJIT HPC Initiative and collaborate with NJIT student organizations including **Association for Computing Machinery (ACM)** and **Women in Computing Society (WiCS)** to organize technical events and expand the campus developer community.
- Organized **14+ technical workshops, hackathons, and Capture the Flag (CTF) competitions** focused on GitHub, artificial intelligence, and software development. ([Event Portfolio](#))

Campus Strategist

September 2024 - December 2024

Perplexity AI

- Led campus outreach and marketing initiatives to increase awareness and adoption of Perplexity AI among students.
- Collaborated with **WiCS** and other student organizations to organize campus activations, AI demonstrations, and community engagement events.
- Worked closely with the Perplexity leadership team and fellow Campus Strategists to gather user feedback, promote AI literacy, and expand the student AI community.

Student Mentor

February 2023 - December 2024

New Jersey Institute of Technology

- Mentored admitted graduate students by connecting with them and answering their questions.
- Organizing events on campus to spread awareness about Parallel Computing being a viable research domain alongside Chapel Programming Language.

HOBBIES

Photography, Philately, CrossFit, Numismatics, Story Writing, Food Blogging.

LANGUAGES

English, Hindi, Sindhi, Marathi, Sanskrit.